

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Name	RAVURU PRANATHI
Register Number	192525076

MODEL PRACTICAL EXAMINATION – SET 4

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application to run a C program to display the student information using any Cloud Service Provider to demonstrate SaaS, with the template of Student Name, Register No., Address, Phone, Age, Courses, Grades and Attendance Report, Progress Report, Semester Mark Sheet forms.
2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.
3. Create a Clone of a VM and test it by loading the previous version/cloned VM.
4. Install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

1. Create a simple cloud software application to run a C program to display the student information using any Cloud Service Provider to demonstrate SaaS, with the template of Student Name, Register No., Address, Phone, Age, Courses, Grades and Attendance Report, Progress Report, Semester Mark Sheet forms.

ANS:

AIM:

To Create a simple cloud software application for Student Information using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step 1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step 4: Enter application name as Student Information

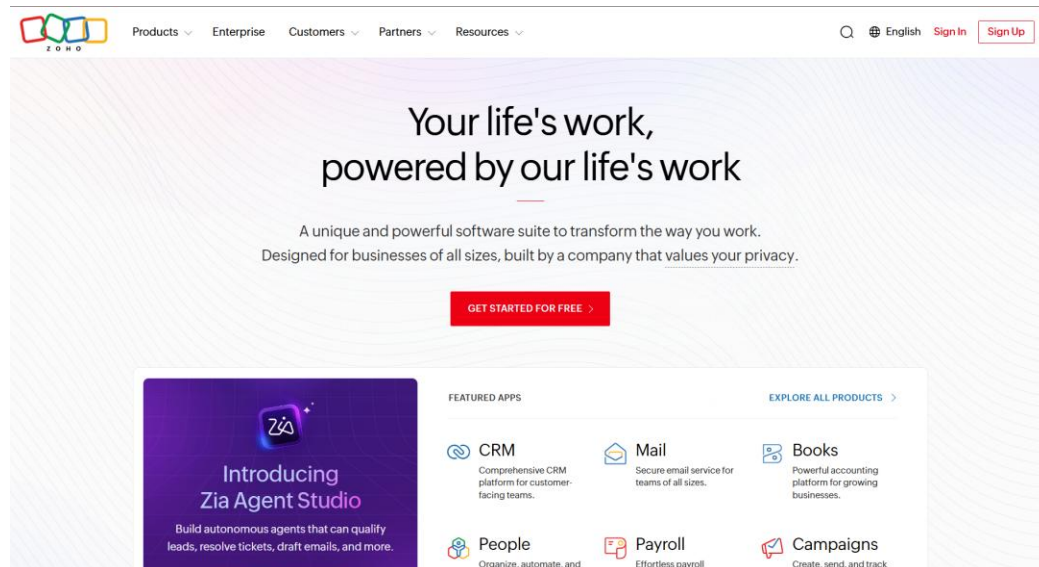
step 5: Created new application as Student Information

step 6: Select one form

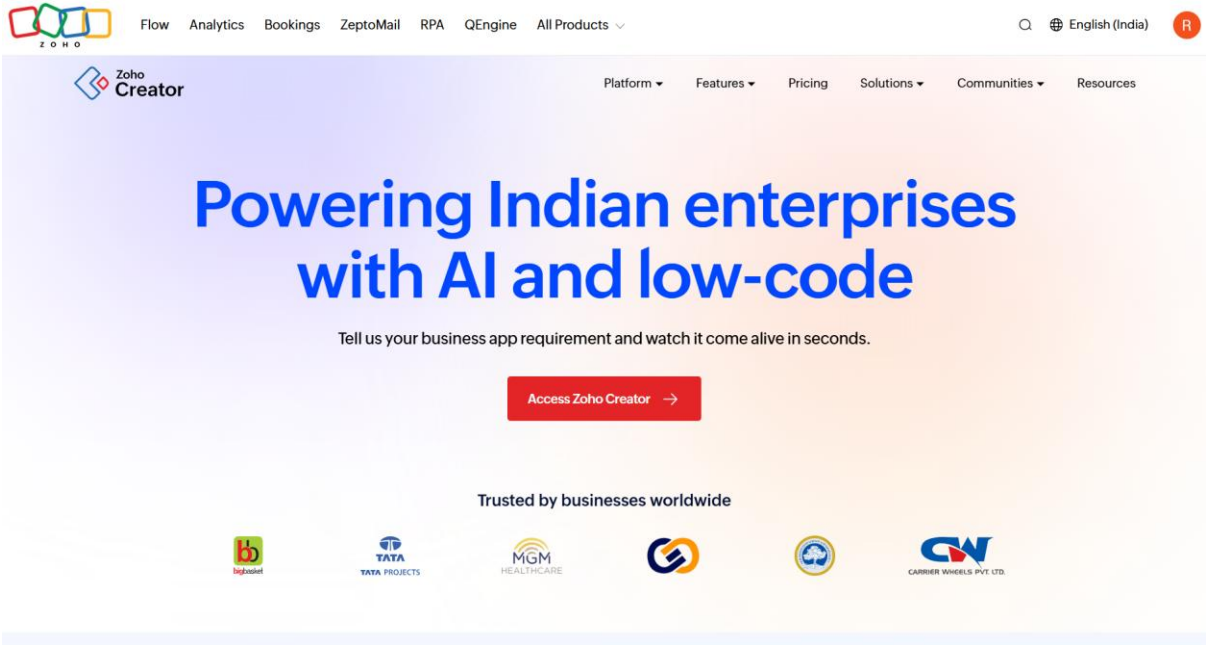
step 7: The software has been created.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

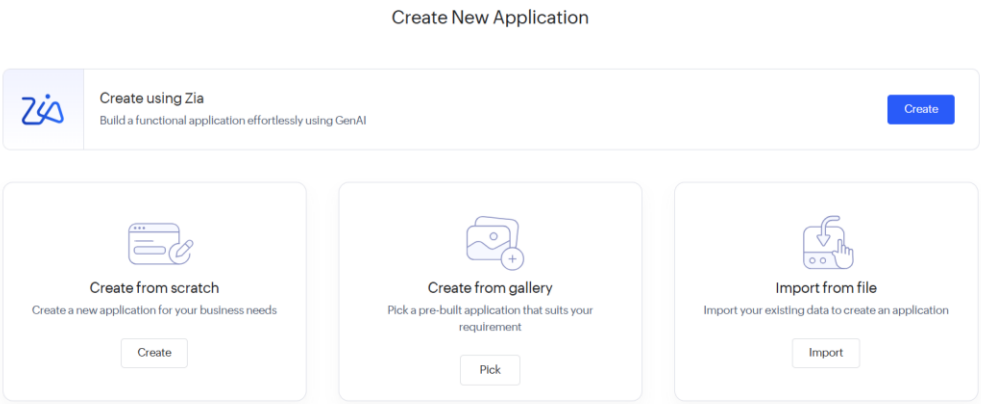


STEP 2: LOGINTO THE ZOHO.COM



STEP 3: CREATE FROM SCRATCH

×



STEP 4: ENTER APPLICATION NAME



Create from scratch

Application Name *

☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION

STUDENT INFORMATION ...

+

Design Workflow Settings

Upgrade

Access this application

STUDENT INFORMATIO...

Start building your application by creating a form

Create New Form

STEP 6: SELECT ONE FORM

How would you like to create your form?

On my own

+

From scratch

↕

Import with data

Zia

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 +
View More

Using an Integrated Datasource

qb

cloud

link

red

color

...

STEP 7: CREATE STUDENT DETAILS FORM

Student Information System
Student Details

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Radio

Multi Select

Student Name

Register Number

Age

Gender

Address

Phone Number

Email

Course

Department

Semester

Field Properties

Field name

Student Name

Field link name

Student_Name

View Field References

Validation

☐ Mandatory

☐ No duplicate values

Initial value

Initial value

Character maximum

255

Data Privacy

☐ Contains personal data(PII)

Data Security

☐ Encrypt data

☐ Contains health info(ePHI)

STEP 10: CREATE PROGRESS REPORT FORM

Student Information System

Progress Report

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Time

Drop Down

Radio

Multi Select

Student Details

Attendance

Average Marks

Grade

Progress Status

Remarks

26

Field Properties

Field name

Student Details

Field link name

Student_Details

View Field References

Display Fields

Student Information System → Student Details

Student Name

+

-

Display type

Drop Down

Validation

Mandatory

Choices

Allow new entries

Alphabetical Order

STEP 11: CREATE WORKFLOW

Form workflows

Schedules

Approvals

Payments

Blueprints

Batch workflows

Functions

Search for Workflows or Acti...

New Workflow

Run actions while forms are being filled out or while records are being edited or deleted.

Create Workflow

See how it works

Here is your Smart Chat (Ctrl+Space)

Trial expires in 6 days | Upgrade

Get started faster!

Help

STEP 12: CREATE RUN WORKFLOW ON ANY EVENT IN THE FORM

Run workflow on any event in the form

Choose form

Progress Report

Run when a record is
(Record Event)

☐ Created ☐ Edited ☒ Created or Edited ☐ Deleted

When to trigger workflow
(Form Event)

User input of a field

Choose Field

Student Details

Name the workflow

Attendance and Average

Create Workflow

STEP 13: WORKFLOWS CREATED FOR STUDENT INFORMATION SYSTEM AND ACCESS THE APPLICATION

Student Information Sys...
Workflow

DesignWorkflowSettings

UpgradeAccess this application

Form workflowsSchedulesApprovalsPaymentsBlueprintsBatch workflowsFunctions

Search for Workflows or Acti...New Workflow

Search for Forms

Forms with workflows

Attendance ReportProgress ReportSemester Mark Sheet

Forms without workflows

Student Details

Workflow Name	Details	Status	Created On
Created or Edited > User Input of Student Details			
Attendance and Average	1 Action	Enabled	29-Jul-2026

STEP 14: IT OPENS THE STUDENT DETAILS FORM AND THEN FILL THE FORM AND CLICK SUBMIT

Student Information System

Student Details

Student Name: Roopa

Register Number: 193765512

Age: 19

Gender: Female

Address: Nellore

Phone Number: +91 34345567678

Email: roopa@gmail.com

Course: B.Sc

Department: CSE

Semester: Semester 2

Submit

STEP 15: IT SAVES THE DETAILS OF STUDENT IN REPORT

Student Information System

Student Details Report

Student Name	Register Number	Age	Gender	Address	Email	Phone Number
Roopa	193765512	19	Female	Nellore	roopa@gmail.com	+9134345567678
Sonu	19237264	21	Male	Nellore	sonu@gmail.com	+913245678675432

Showing 2 of 2

STEP 16: AND FILL THE FORM OF ATTENDANCE REPORT AND THEN CLICK SUBMIT

SIS Student Information System

Student Details >

Attendance Report ▾

Attendance Report

Attendance Report Report

Semester Mark Sheet >

Progress Report >

V Venakta Rao ...

Attendance Report

Student Details

Roopa

Total Classes

77

Present

20

Absent

57

Attendance Percentage

25.97

Attendance Status

Shortage

Submit

STEP 17: IT SAVES THE DETAILS OF ATTENDANCE IN REPORT IT ALSO CALCULATES ABSENT,PERCENTAGE AND STATUS

SIS Student Information System

Student Details >

Attendance Report ▾

Attendance Report

Attendance Report Report

Semester Mark Sheet >

Progress Report >

V Venakta Rao ...

Attendance Report Report

Student Details ▾

Total Classes ▾

Present ▾

Absent ▾

Attendance Percentage ▾

Attendance Status ▾

Roopa	77	20	57	25.97	Shortage
Sonu	78	70	8	89.74	Eligible

Showing 2 of 2

STEP 18: FILL THE SEMESTER MARKSHEET FORM AND CLICK SUBMIT

Student Information System

Trial expires in 12 days [Upgrade](#) [Edit this application](#) [Help](#)

Student Information System

- Student Details
- Attendance Report
- Semester Mark Sheet
 - Semester Mark Sheet**
 - Semester Mark Sheet Report
- Progress Report

Semester Mark Sheet

Mark 1: 87

Subject 2: chemistry

Mark 2: 52

Subject 3: maths

Mark 3: 43

Subject 4: Computing

Mark 4: 94

Subject 5: C++

Mark 5: 85

Total: 361

Average: 72.20

Grade: A

[Submit](#)

STEP 19: IT SAVES THE DETAILS OF SEMESTER MARKSHEET IN REPORT...CALCULATES AVG,TOTAL,GRADE

Student Information System

Trial expires in 12 days [Upgrade](#) [Edit this application](#) [Help](#)

Student Information System

- Student Details
- Attendance Report
- Semester Mark Sheet
 - Semester Mark Sheet
 - Semester Mark Sheet Report**
- Progress Report

Semester Mark Sheet Report

Showing 2 of 2

☐	Student Details	Subject 1	Mark 1	Subject 2	Mark 2	Subject 3	Mark 3	Total	Average	Grade
☐	Roopa	Physics	87	chemistry	52	maths	43	361	72.20	A
☐	Sonu	Mathematics	89	Physics	90	Chemistry	95	454	90.80	O

STEP 20: FILL THE PROGRESS REPORT FORM AND CLICK SUBMIT

Student Information System

Trial expires in 12 days Upgrade Edit this application Help

Student Information System

Student Details >

Attendance Report >

Semester Mark Sheet >

Progress Report ▾

Progress Report

Progress Report Report

Progress Report

Student Details

Roopa

Attendance

25

Average Marks

72.20

Grade

A

Progress Status

Poor

Remarks

Attend Remedial Classes

Submit

STEP 21: IT SAVES THE DETAILS OF PROGRESS REPORT WHEN WE ENTER STUDENT NAME

Student Information System

Trial expires in 12 days Upgrade Edit this application Help

Student Information System

Student Details >

Attendance Report >

Semester Mark Sheet >

Progress Report ▾

Progress Report

Progress Report Report

Progress Report Report

Student Details ▾

Attendance ▾

Average Marks ▾

Grade ▾

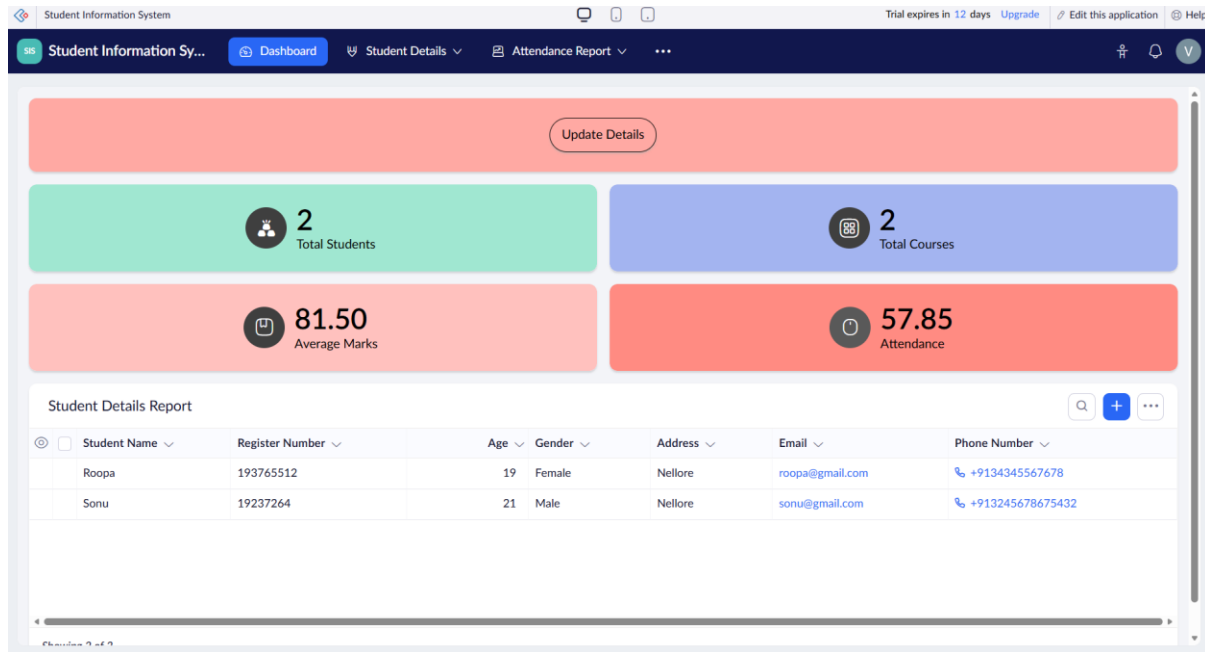
Progress Status ▾

Remarks ▾

Roopa	25	72.20	A	Poor	Attend Remedial Classes
Sonu	89	90.80	S	Excellent	Outstanding Performance

Showing 2 of 2

STEP 22: DASHBOARD



Result

A cloud-based C application was successfully created and executed using an online cloud compiler. The program displayed the student information, courses, grades, attendance, progress report, and semester marks, successfully demonstrating the SaaS (Software as a Service) model.

2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.

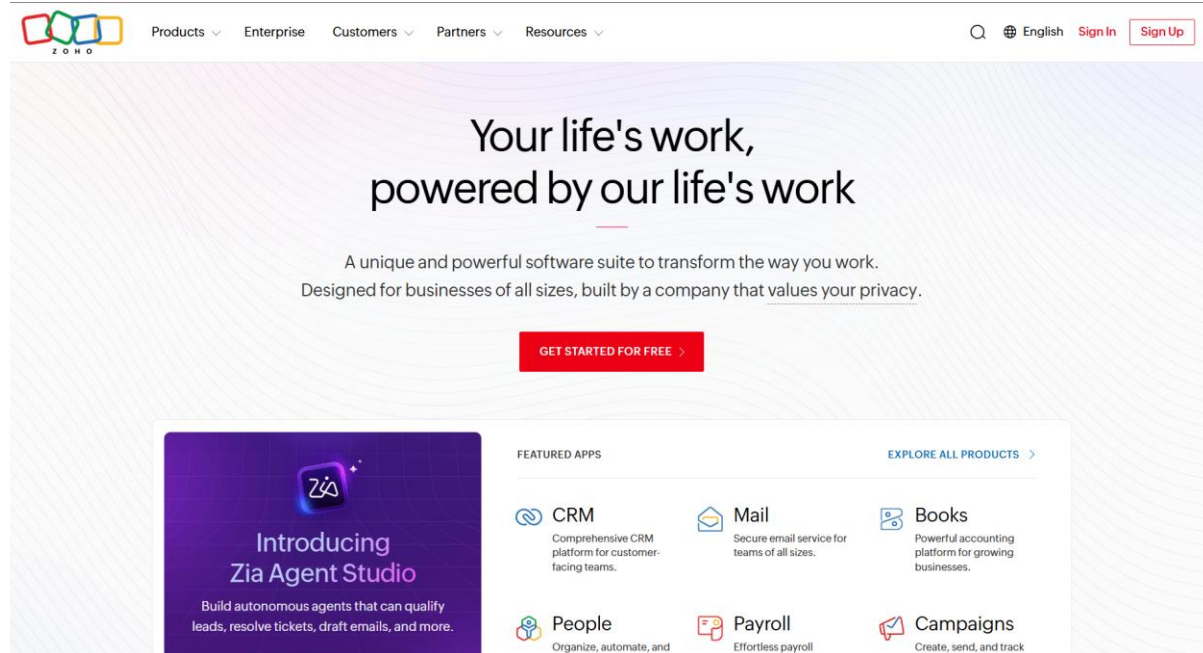
ANS:

AIM: To Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

- step 1: Go to zoho.com.
- step 2: Log into the zoho.com.
- step 3: Select one application step.
- step 4: Enter application name as Mark Sheet Processing System.
- step 5: Created new application as Mark Sheet Processing System.
- step 6: Select one form
- step 7: The software has been created.

IMPLEMENTATION: STEP1: GOTO ZOHO.COM



The Zoho homepage features a clean, modern design with a light blue and white color scheme. At the top, there is a navigation bar with the Zoho logo on the left and links for Products, Enterprise, Customers, Partners, and Resources in the center. On the right, there are links for English, Sign In, and Sign Up. The main headline reads "Your life's work, powered by our life's work" in a large, bold, black font. Below this, a sub-headline states: "A unique and powerful software suite to transform the way you work. Designed for businesses of all sizes, built by a company that values your privacy." A prominent red button with the text "GET STARTED FOR FREE" is centered below the sub-headline. The lower section of the page is divided into two main areas. On the left, there is a purple card titled "Introducing Zia Agent Studio" with a small Zia icon. The text on the card says: "Build autonomous agents that can qualify leads, resolve tickets, draft emails, and more." On the right, there is a section titled "FEATURED APPS" with a link to "EXPLORE ALL PRODUCTS". This section displays six app icons with their names and brief descriptions: CRM (Comprehensive CRM platform for customer-facing teams), Mail (Secure email service for teams of all sizes), Books (Powerful accounting platform for growing businesses), People (Organize, automate, and), Payroll (Effortless payroll), and Campaigns (Create, send, and track).

Products ▾ Enterprise Customers ▾ Partners ▾ Resources ▾

Q English Sign In Sign Up

Your life's work, powered by our life's work

A unique and powerful software suite to transform the way you work.
Designed for businesses of all sizes, built by a company that values your privacy.

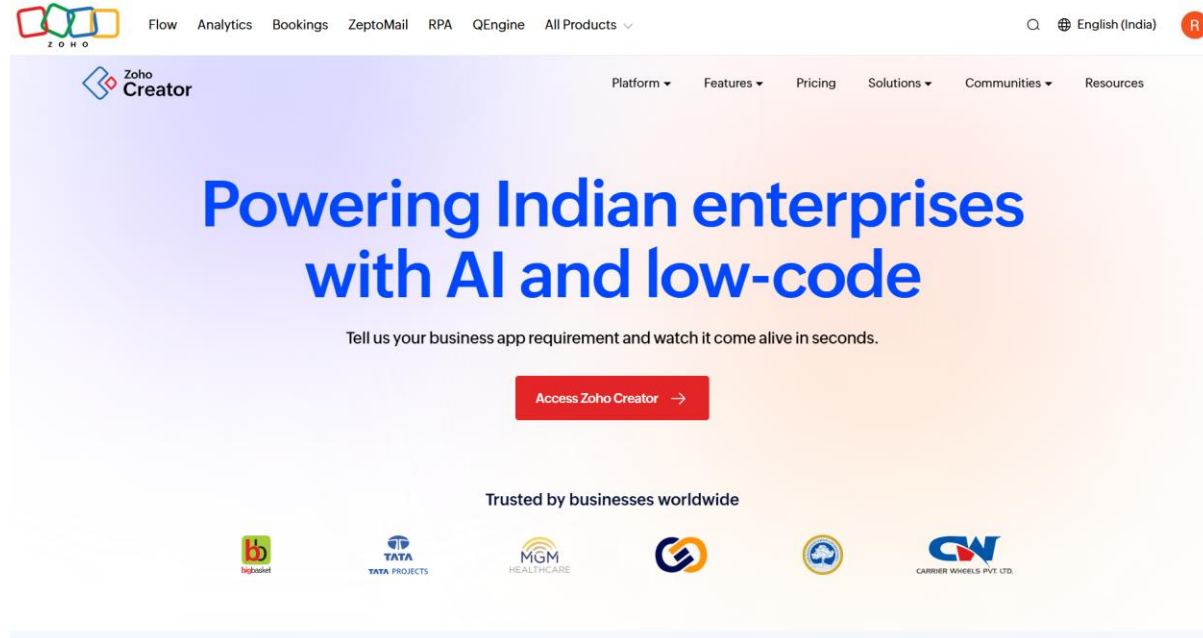
GET STARTED FOR FREE >

Introducing Zia Agent Studio
Build autonomous agents that can qualify leads, resolve tickets, draft emails, and more.

FEATURED APPS EXPLORE ALL PRODUCTS >

- CRM**
Comprehensive CRM platform for customer-facing teams.
- Mail**
Secure email service for teams of all sizes.
- Books**
Powerful accounting platform for growing businesses.
- People**
Organize, automate, and
- Payroll**
Effortless payroll
- Campaigns**
Create, send, and track

STEP 2: LOGINTO THE ZOHO.COM



The Zoho Creator homepage has a light blue and white background. The top navigation bar includes the Zoho logo, links for Flow, Analytics, Bookings, ZeptoMail, RPA, QEngine, and All Products, and links for English (India) and a user profile icon. Below the navigation bar, the Zoho Creator logo is on the left, and a horizontal menu with links for Platform, Features, Pricing, Solutions, Communities, and Resources is on the right. The main headline is "Powering Indian enterprises with AI and low-code" in a large, bold, blue font. Below this, a sub-headline reads: "Tell us your business app requirement and watch it come alive in seconds." A red button with the text "Access Zoho Creator" and a right arrow is centered below the sub-headline. At the bottom, there is a section titled "Trusted by businesses worldwide" with a row of logos for various companies: Injushel, TATA PROJECTS, MGM HEALTHCARE, a company with a blue and orange logo, a company with a blue and yellow logo, and CARRIER WHEELS PVT. LTD.

Flow Analytics Bookings ZeptoMail RPA QEngine All Products ▾

Q English (India) R

Zoho Creator Platform ▾ Features ▾ Pricing Solutions ▾ Communities ▾ Resources

Powering Indian enterprises with AI and low-code

Tell us your business app requirement and watch it come alive in seconds.

Access Zoho Creator →

Trusted by businesses worldwide

Injushel TATA PROJECTS MGM HEALTHCARE [Logo] [Logo] CARRIER WHEELS PVT. LTD.

STEP 3: CREATE FROM SCRATCH

×

Create New Application



Create using Zia
Build a functional application effortlessly using GenAI

Create



Create from scratch
Create a new application for your business needs

Create



Create from gallery
Pick a pre-built application that suits your requirement

Pick



Import from file
Import your existing data to create an application

Import

STEP 4: ENTER APPLICATION NAME

Create from scratch

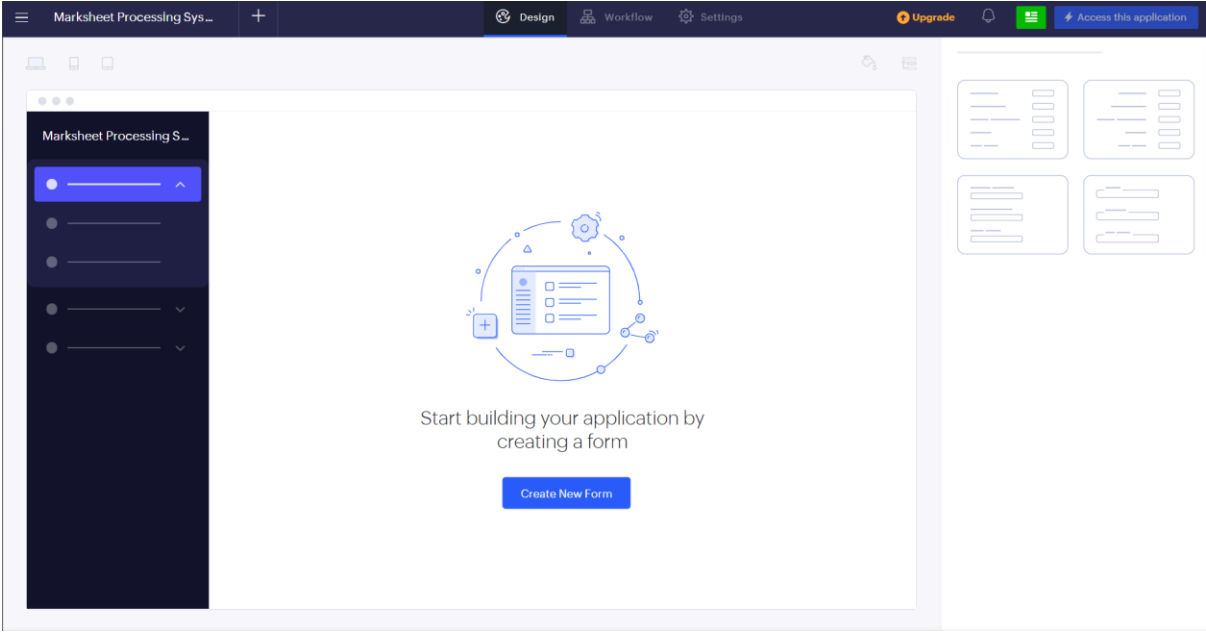
Application Name *

Marksheet Processing System

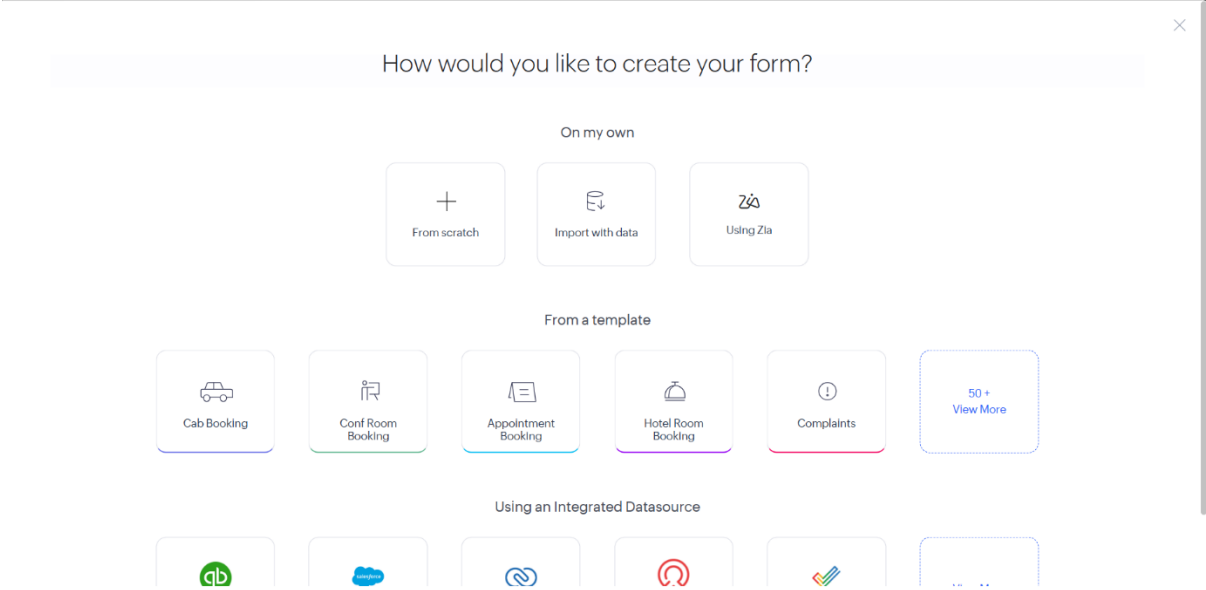
☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: CREATE STUDENT DETAILS FORM

Mark Sheet Processing system

Student Details

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Radio

Multi Select

Student ID

Student Name

Register Number

Department

Year

Semester

Section

Email

Phone Number

Field Properties

Field name

Student ID

Field link name

Student_ID

View Field References

Validation

Start Index

1

Appearance

Field type

Auto Number

STEP 8: CREATE MARKS ENTRY FORM

Mark Sheet Processing system

Marks Entry

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Radio

Multi Select

Student Details

Subject 1

Mark 1

Subject 2

Mark 2

Subject 3

Mark 3

Subject 4

Mark 4

Subject 5

Mark 5

Average

Field Properties

Field name

Student Details

Field link name

Student_Details

View Field References

Display Fields

Mark Sheet Processing system → Student Details

Prefix

First Name

Last Name

Suffix

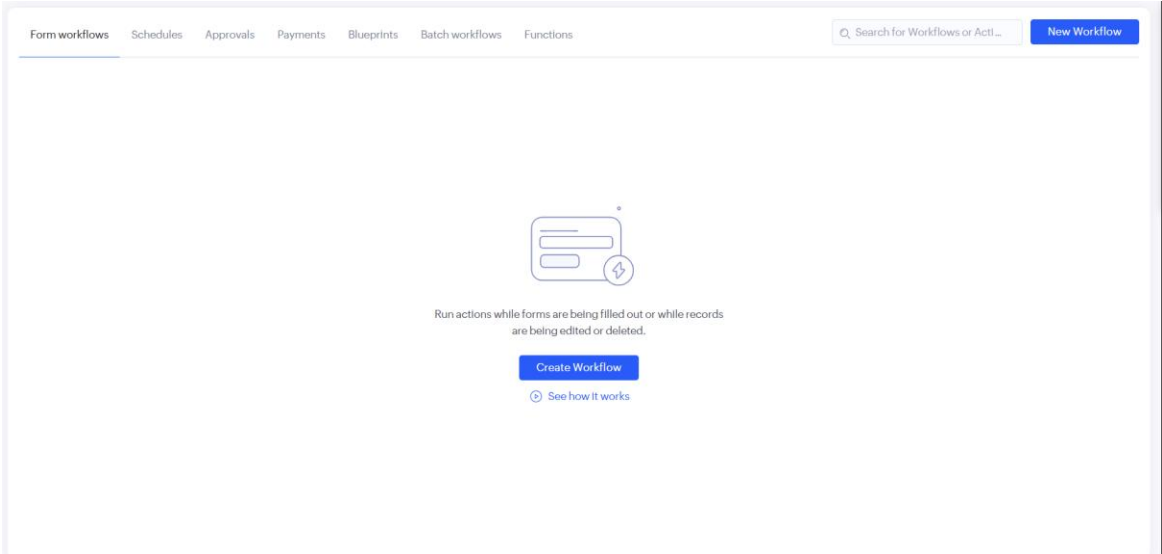
Preview

Prefix First Name Last Name Suffix

Display type

Drop Down

STEP 9: CREATE WORKFLOW



STEP 10: CREATE RUN WORKFLOW ON ANY EVENT IN THE FORM

Run workflow on any event in the form

Choose form

Marks Entry

Run when a record is
(Record Event)

☐ Created ☐ Edited ☒ Created or Edited ☐ Deleted

When to trigger workflow
(Form Event)

User input of a field

Choose Field

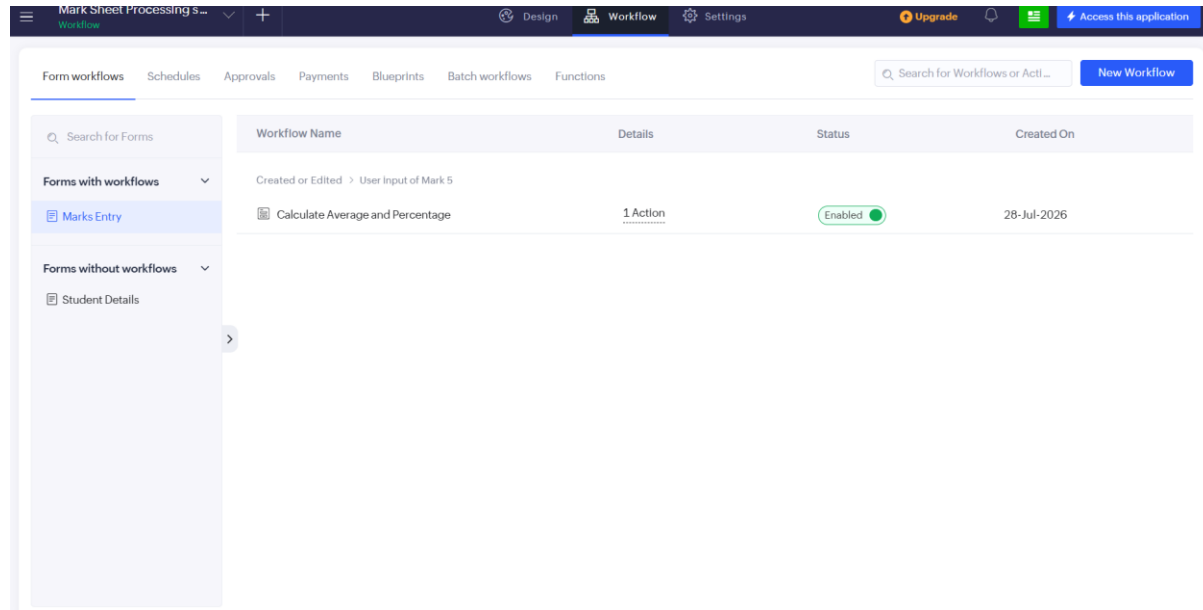
Mark 5

Name the workflow

Calculate Average and Percentage

Create Workflow

STEP 11: WORKFLOWS CREATED FOR MARK PROCESSING SYSTEM AND ACCESS THE APPLICATION



STEP 12: IT OPENS THE STUDENT DETAILS FORM AND THEN FILL THE FORM AND CLICK SUBMIT

The screenshot shows the 'Student Details' form. The left sidebar has 'Student Details' selected. The form fields are filled with the following information:

- Student Name: Swaroopa (First Name), Tata (Last Name)
- Register Number: 193352576
- Department: CSE
- Year: 2nd
- Semester: Semester 2
- Section: A
- Email: swaroopa@gmail.com
- Phone Number: +91 8765432456

A 'Submit' button is located at the bottom of the form.

STEP 13: IT SAVES THE DETAILS OF STUDENT IN REPORT

Mark Sheet Processing system

Student Details Report

Student ID Student Name Register Number Department Year Semester Section

2	Swaroopata Tata	193352576	CSE	2nd	Semester 2	A
1	Sonu Balikana	19252543	CSE	3rd	semester 3	A

Showing 2 of 2

Venakta Rao ...

STEP 14: AND FILL THE FORM OF MARKS ENTRY DETAILS AND THEN CLICK SUBMIT

Mark Sheet Processing system

Marks Entry

Subject 2 Physics

Mark 2 88

Subject 3 Chemistry

Mark 3 92

Subject 4 Cloud

Mark 4 85

Subject 5 Python

Mark 5 95

Average 91.60

Percentage 91.60

Rank #####

Over all Performance Excellent

Submit

Venakta Rao ...

STEP 15: IT SAVES THE DETAILS OF MARKS ENTRY IN REPORT

Mark Sheet Processing system

Trial expires in 14 days Upgrade Edit this application Help

Mark Sheet Processing system

Student Details >

Marks Entry >

Marks Entry Report

Marks Entry Report

Student Details > Subject 1 > Mark 1 > Subject 2 >

... Swaroopa Tata Mathematics 98 Physics

... Sonu Balikana Mathematics 98 Physics

Showing 2 of 2

Student Details Sonu Balikana

Subject 1 Mathematics

Mark 1 98

Subject 2 Physics

Mark 2 93

Subject 3 Chemistry

Mark 3 96

Subject 4 Computing

Mark 4 89

Subject 5 Python

Mark 5 92

Average 93.60

Percentage 93.60

Rank

Over all Performance Excellent

Add a comment

STEP 16: IT AUTOMATICALLY CALCULATES THE AVERAGE, PERCENTAGE, AND OVERALL PERFORMANCE

Mark Sheet Processing system

Trial expires in 14 days Upgrade Edit this application Help

Mark Sheet Processing system

Student Details >

Marks Entry >

Marks Entry Report

Marks Entry Report

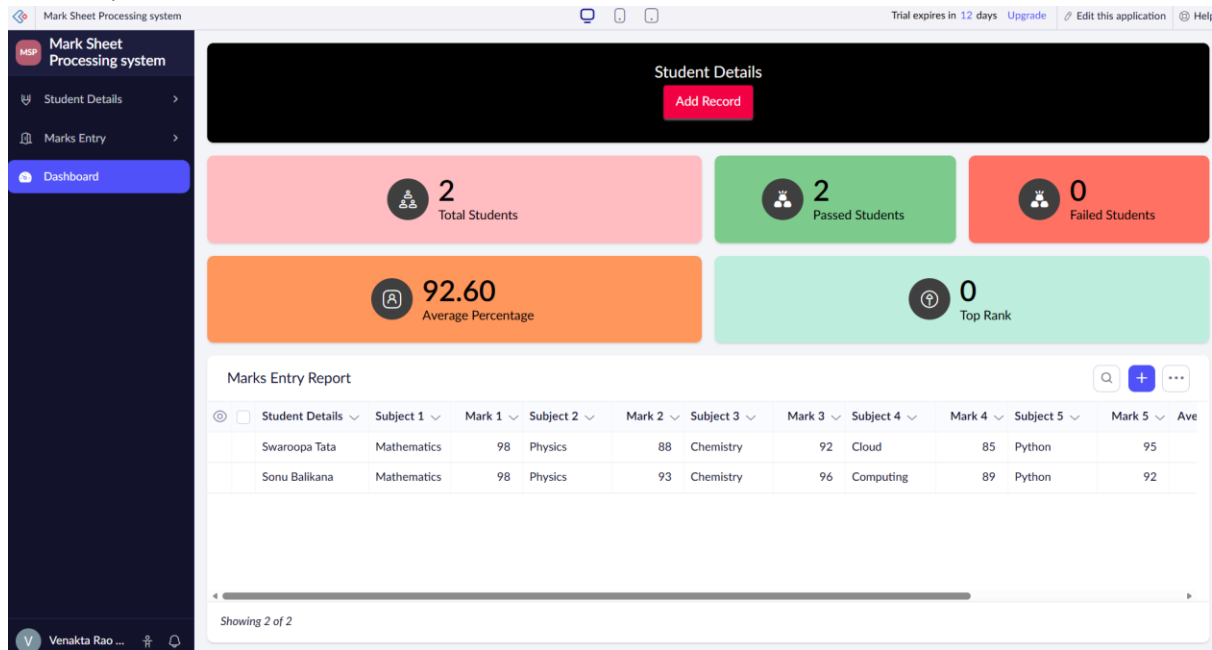
Mark 1 > Subject 2 > Mark 2 > Subject 3 > Mark 3 > Subject 4 > Mark 4 > Subject 5 > Mark 5 > Average > Percentage > Over all Performan...

98 Physics 88 Chemistry 92 Cloud 85 Python 95 91.60 91.60 Excellent

98 Physics 93 Chemistry 96 Computing 89 Python 92 93.60 93.60 Excellent

Showing 2 of 2

STEP 17: DASHBOARD



Result:

A cloud-based **Mark Sheet Processing System** was successfully created using Google Sheets. Student marks were processed to calculate **Total, Average, Percentage, Rank, and Overall Performance** automatically. The application was accessed through the cloud, successfully demonstrating the **SaaS (Software as a Service) model**.

3. Create a Clone of a VM and test it by loading the previous version/cloned VM.

ANS:

Aim:

To create a clone of an existing virtual machine and test the cloned virtual machine using VMware Workstation.

Procedure:

Step 1: Shut down the existing Ubuntu VM completely.

Step 2: Open VMware Workstation and select VM → Manage → Clone.

Step 3: Select the current state of the virtual machine as the cloning source.

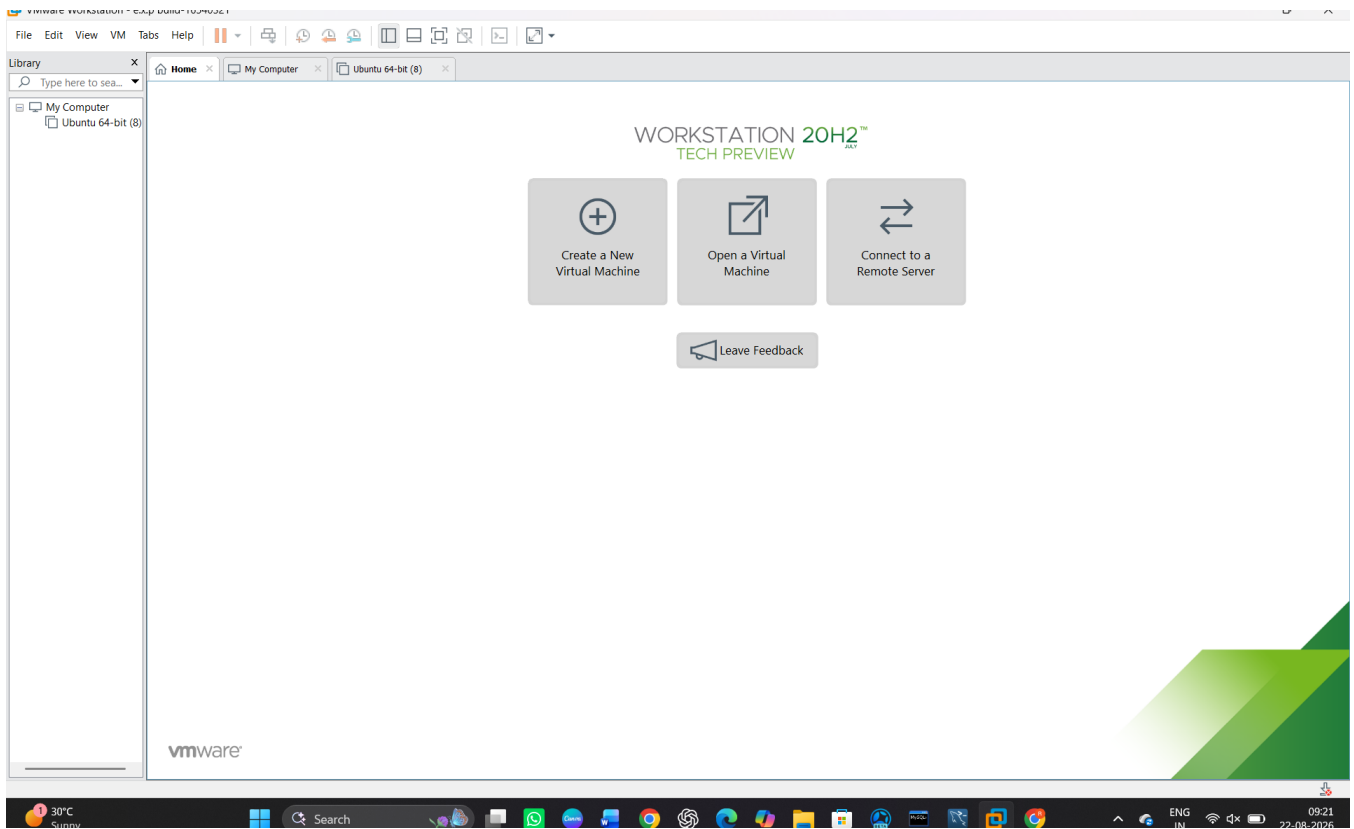
Step 4: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.

Step 5: Complete the cloning process and start the Ubuntu_Clone virtual machine.

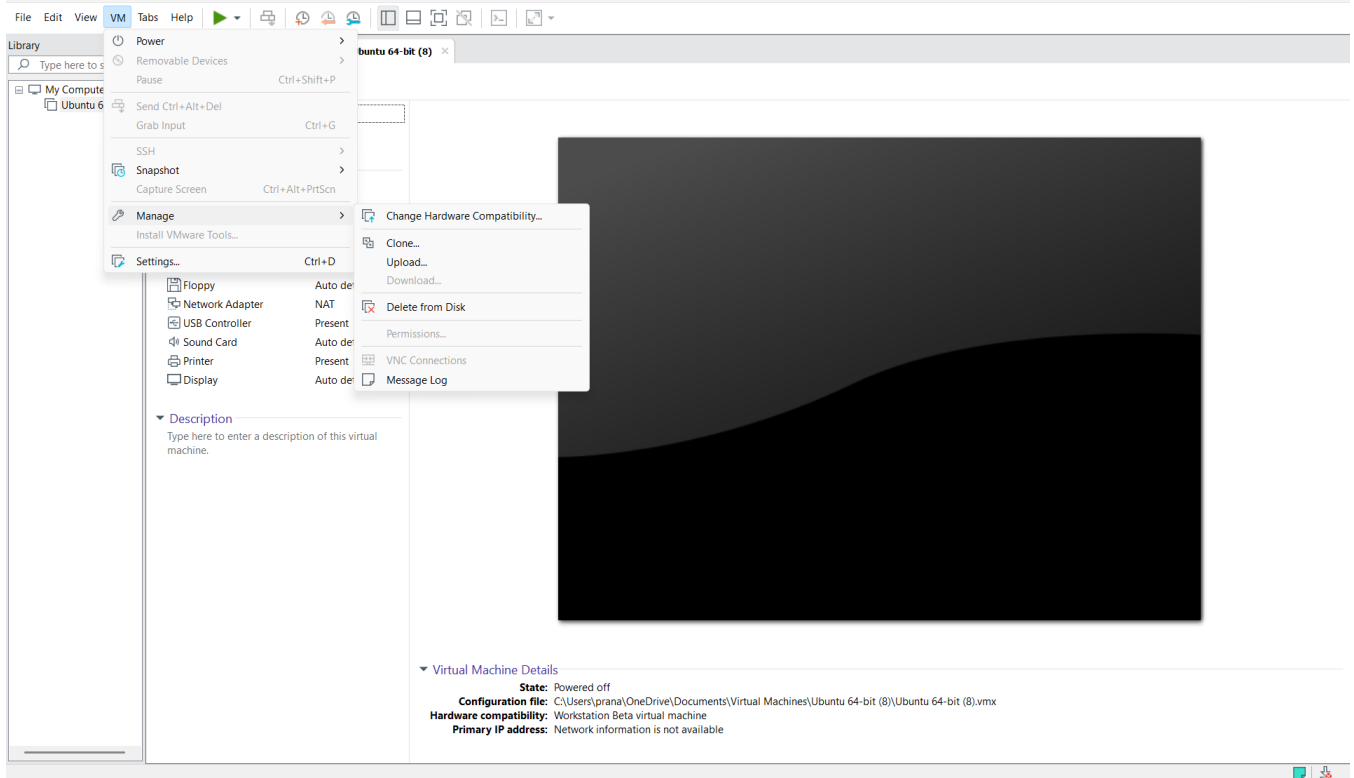
Step 6: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.

IMPLEMENTATION:

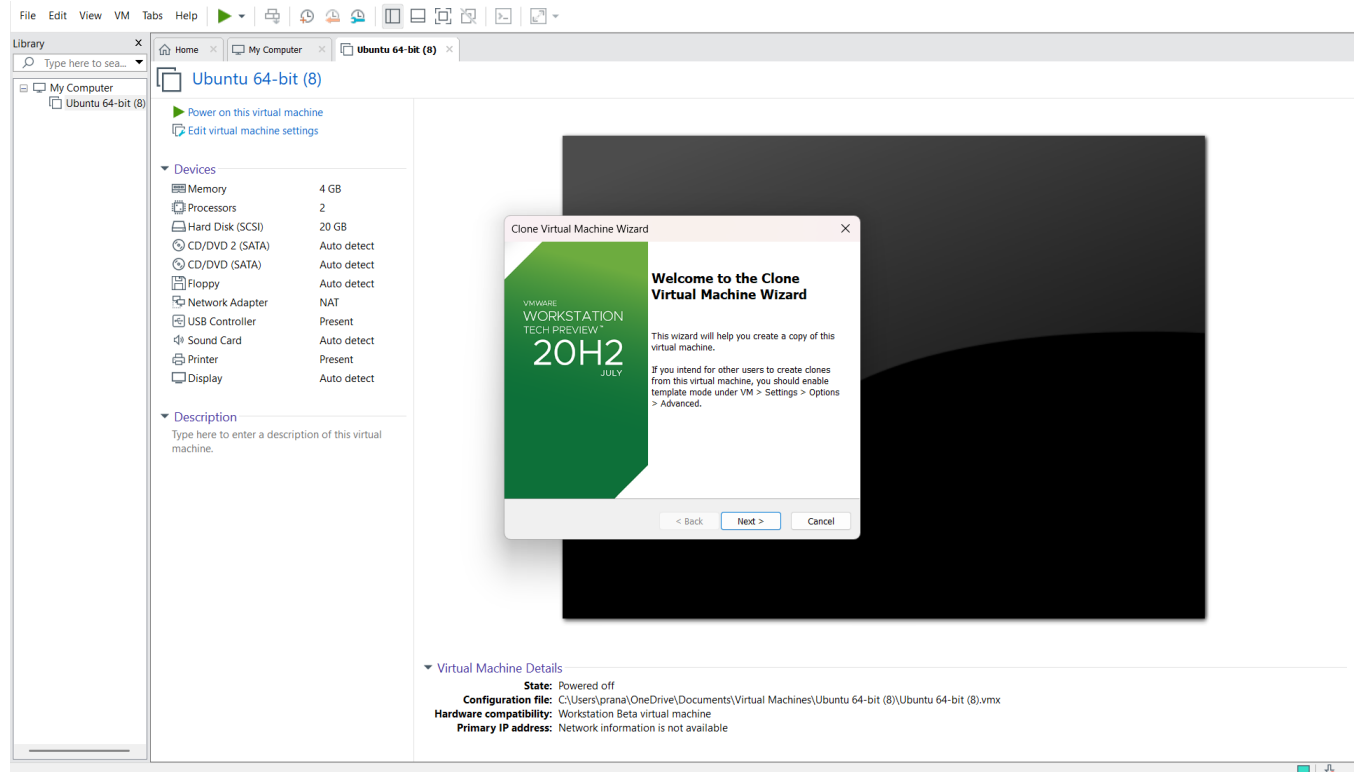
STEP 1: GOTO VMWARE WORKSTATION



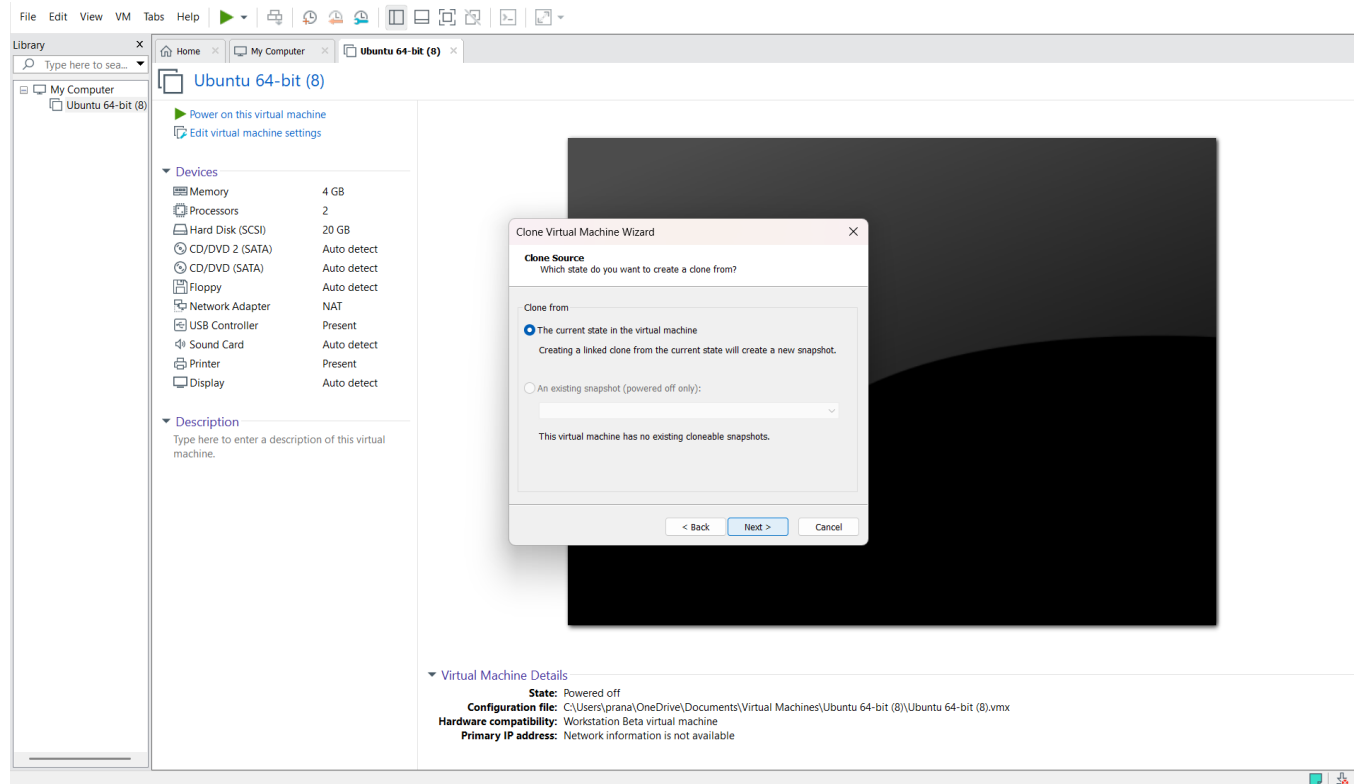
Step 2: Open VMware Workstation and select VM → Manage → Clone.



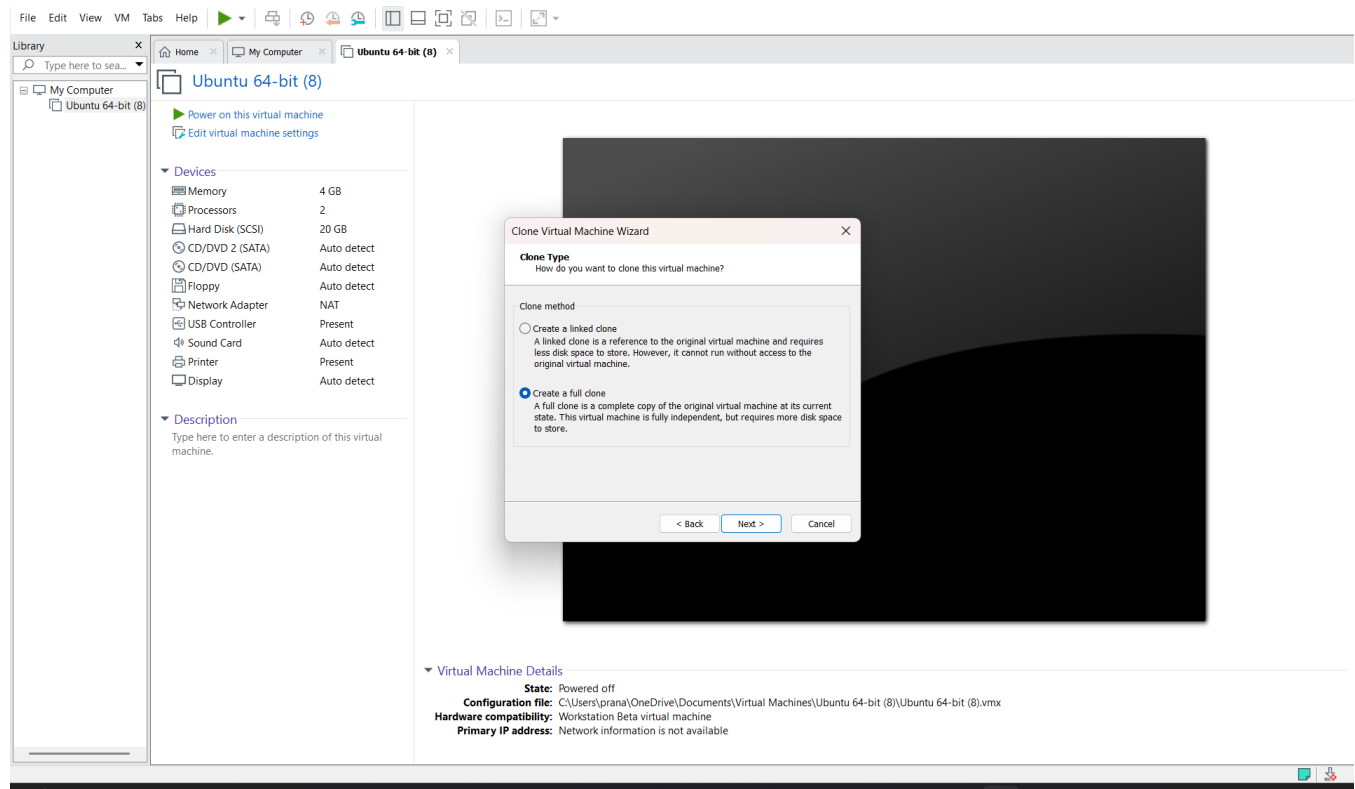
STEP 3: CLICK NEXT



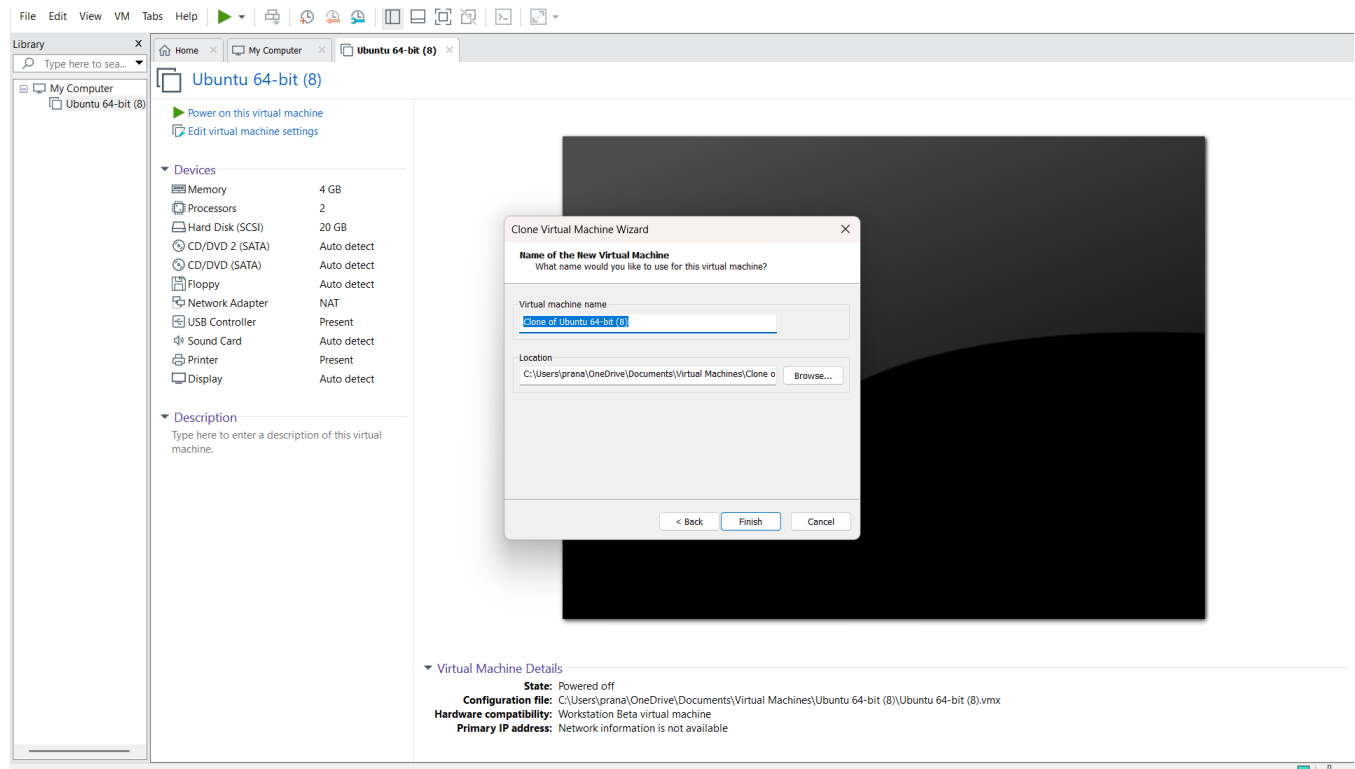
Step 4: Select the current state of the virtual machine as the cloning source.



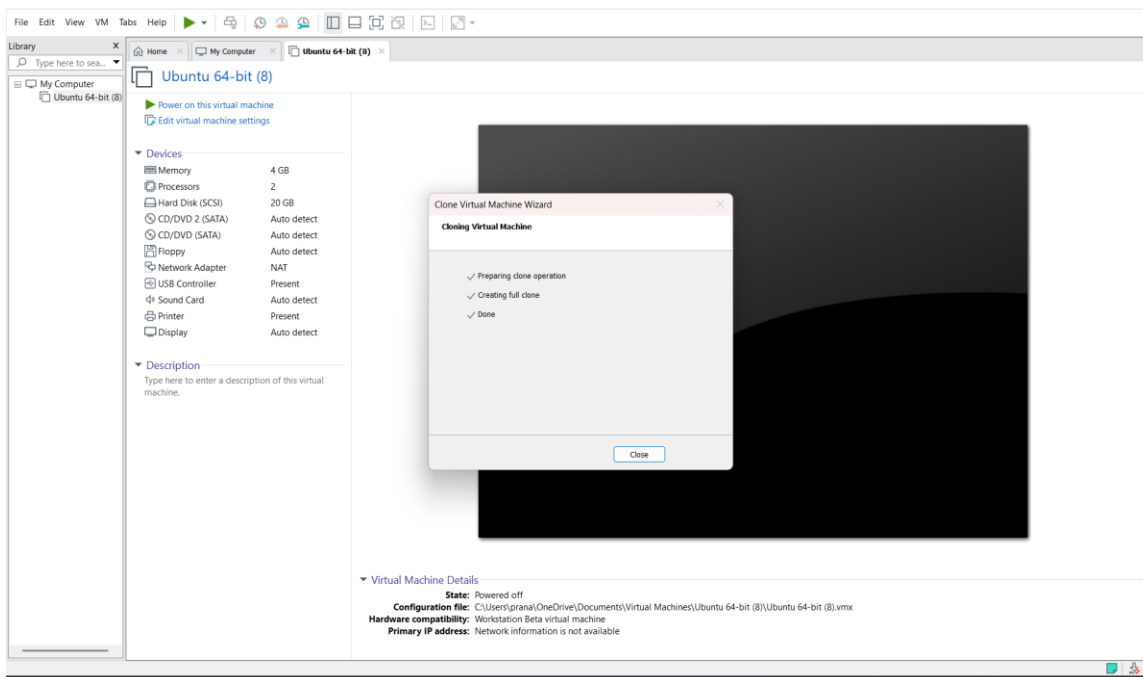
STEP 5: SELECT THE FULL CLONE



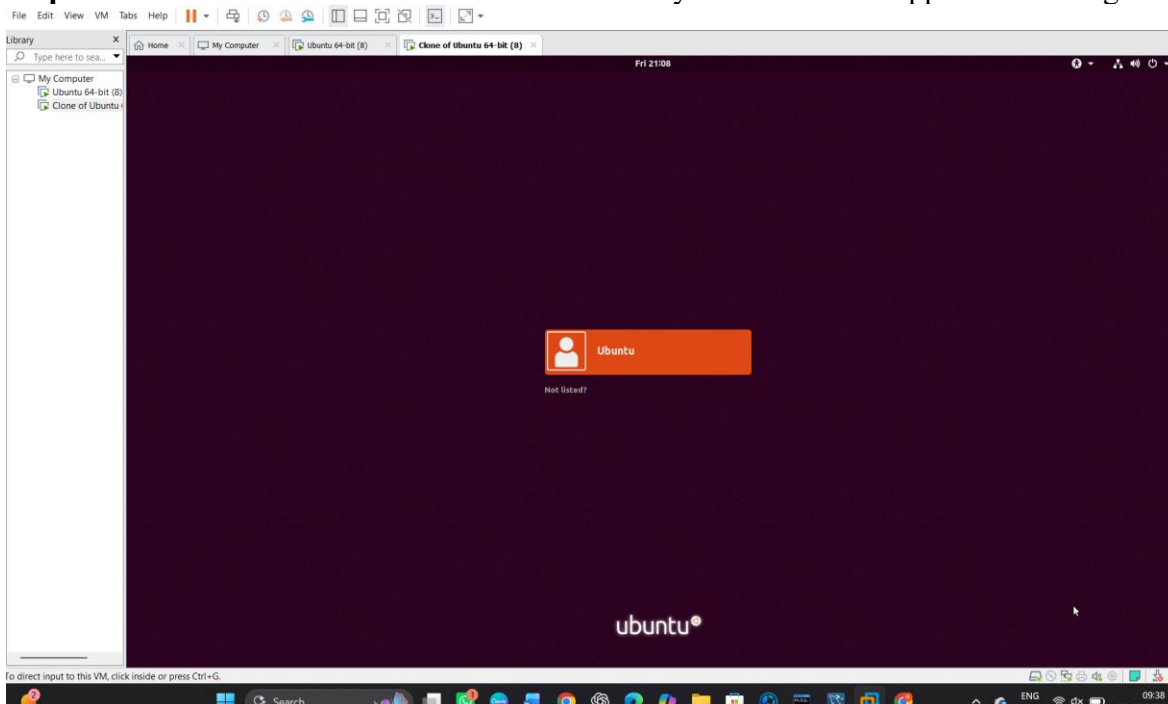
Step 6: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.



Step 7: Complete the cloning process and start the Ubuntu_Clone virtual machine.



Step 8: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.



Result: The Ubuntu virtual machine was successfully cloned using VMware Workstation. The cloned VM was loaded and tested successfully, and it booted and operated correctly, demonstrating successful VM cloning and restoration/loading of the cloned version.

4. Install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

ANS:

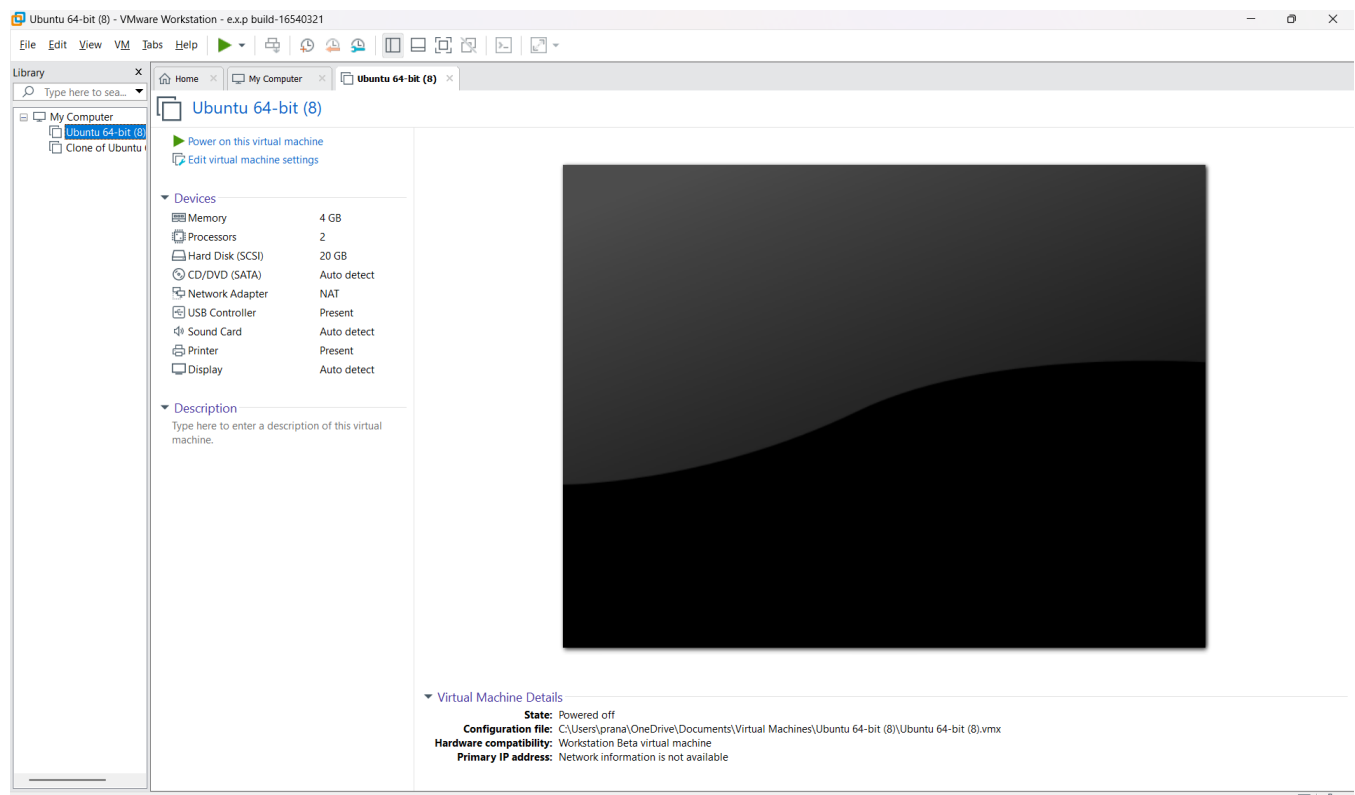
Aim:

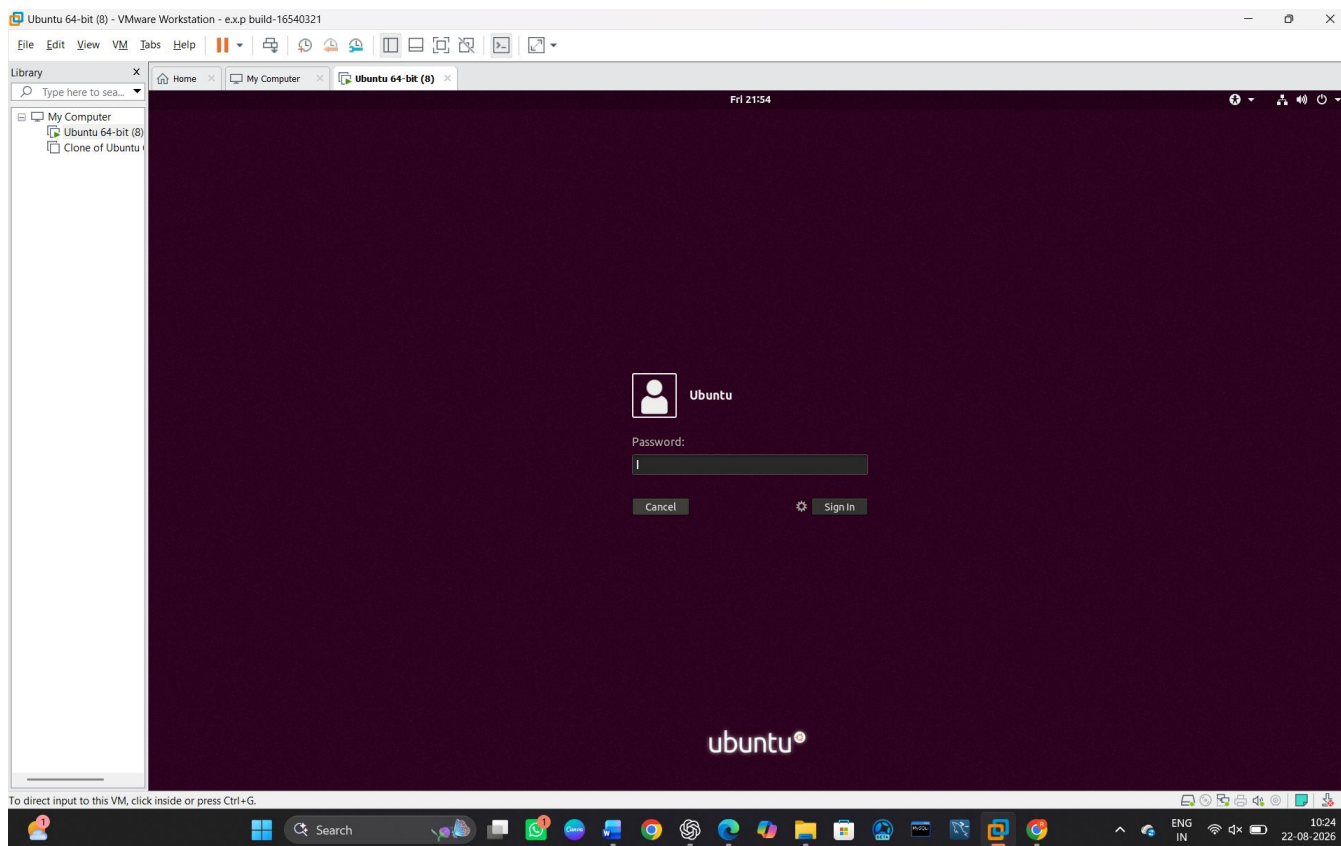
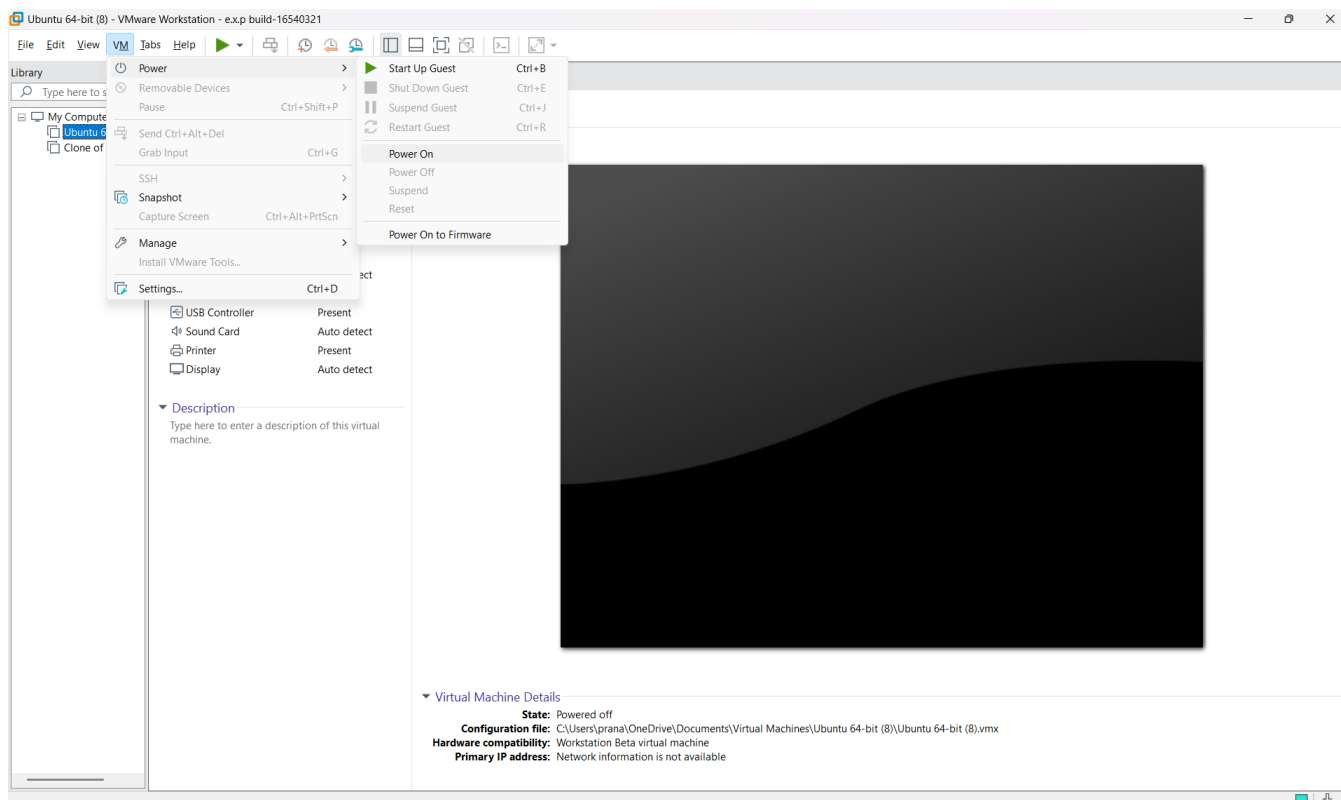
To install a **C compiler (GCC)** in the Ubuntu virtual machine created using **VMware Workstation** and execute simple C programs.

Procedure:

Step 1:

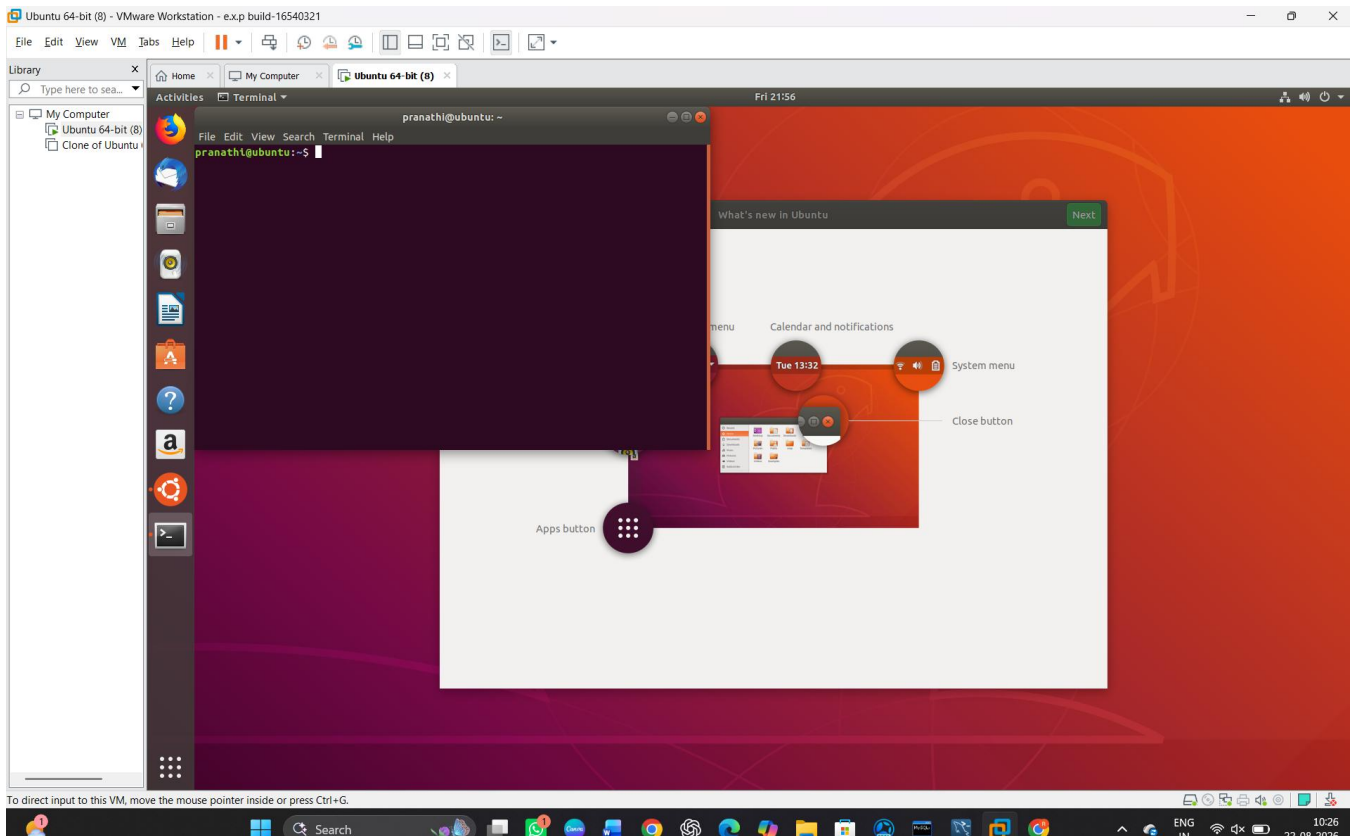
1. Start the Virtual Machine
2. Open VMware Workstation.
3. Select Ubuntu 64-bit (8).
4. Click Power On.
5. Log in to Ubuntu and enter password



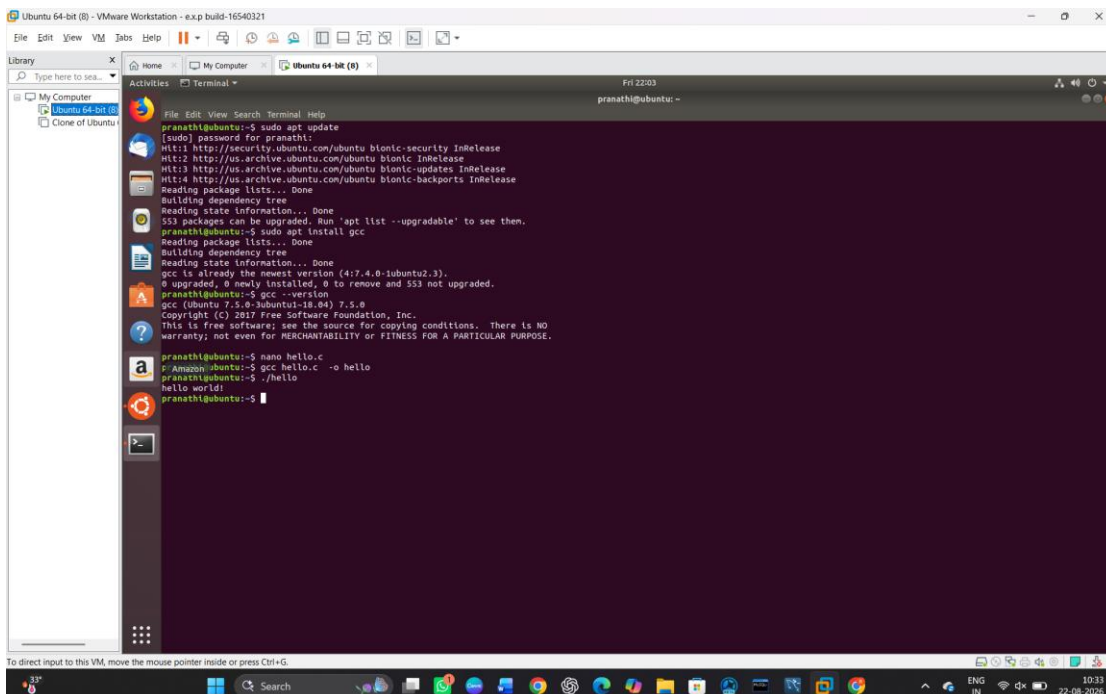


Step 2: Open Terminal

Enter Ctrl + Alt + T

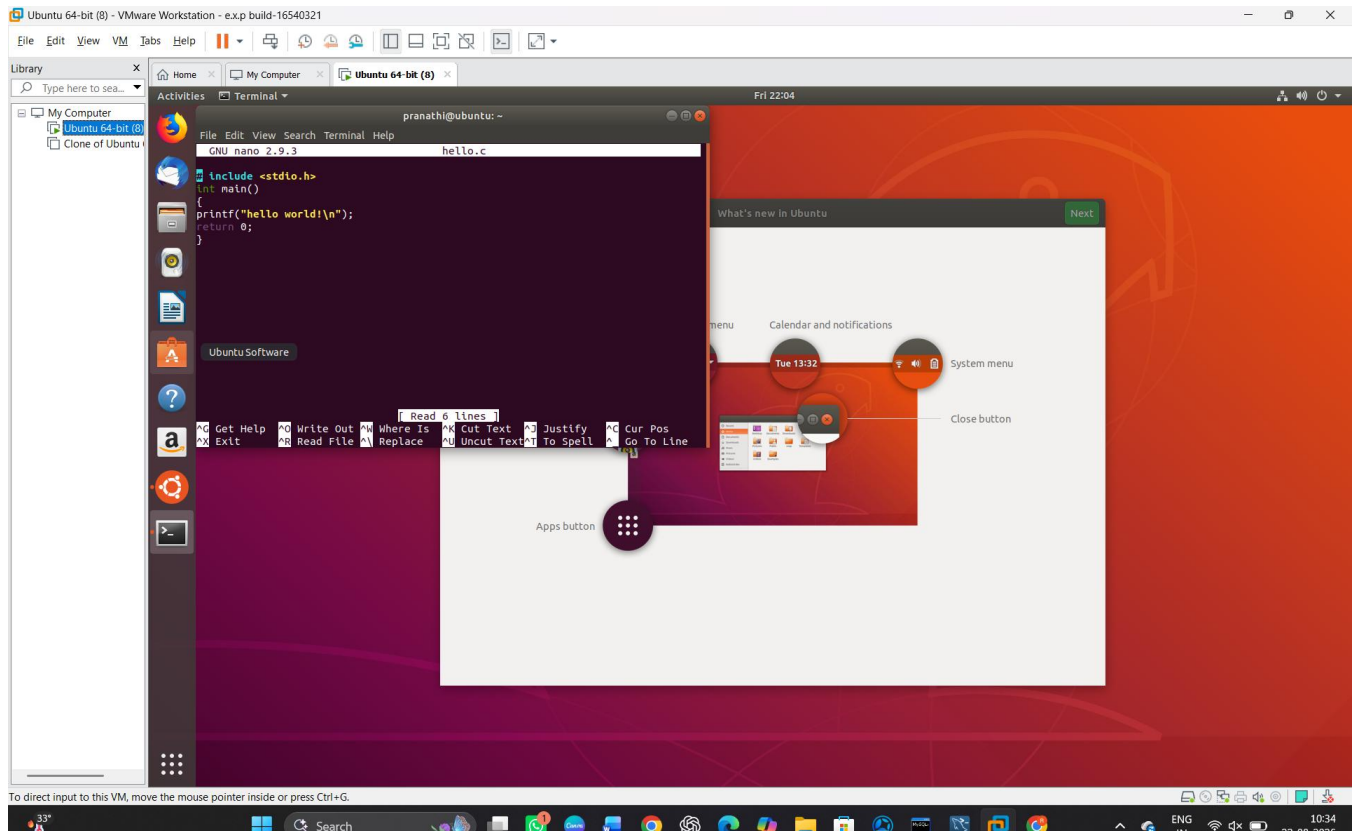


Step 3: Update Package Information



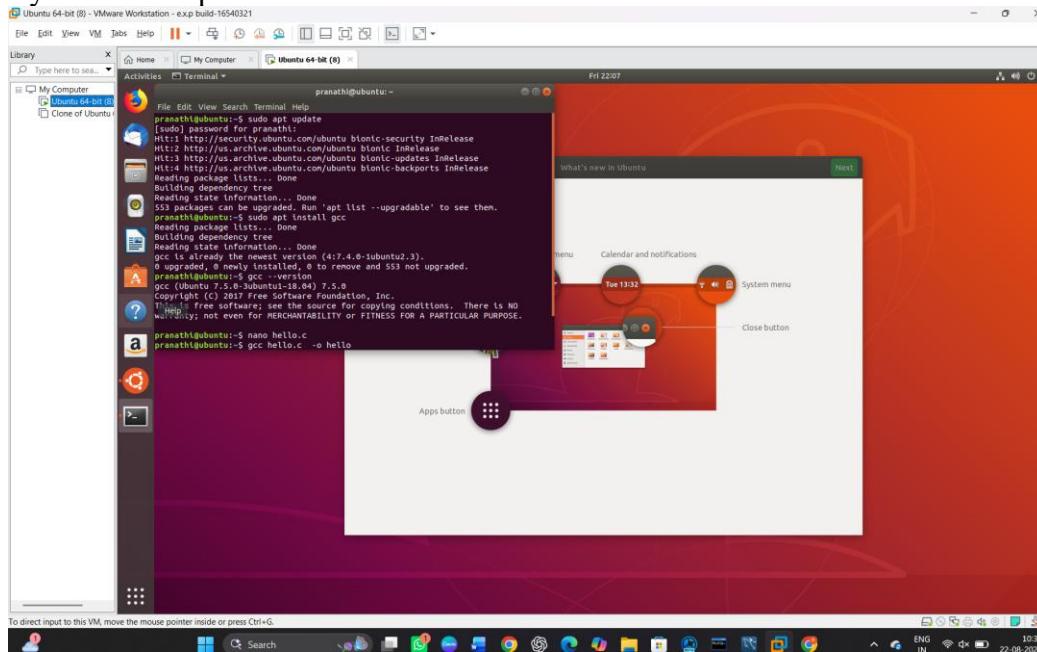
Step 4: Create the C program

Type nano hello..c and press Enter



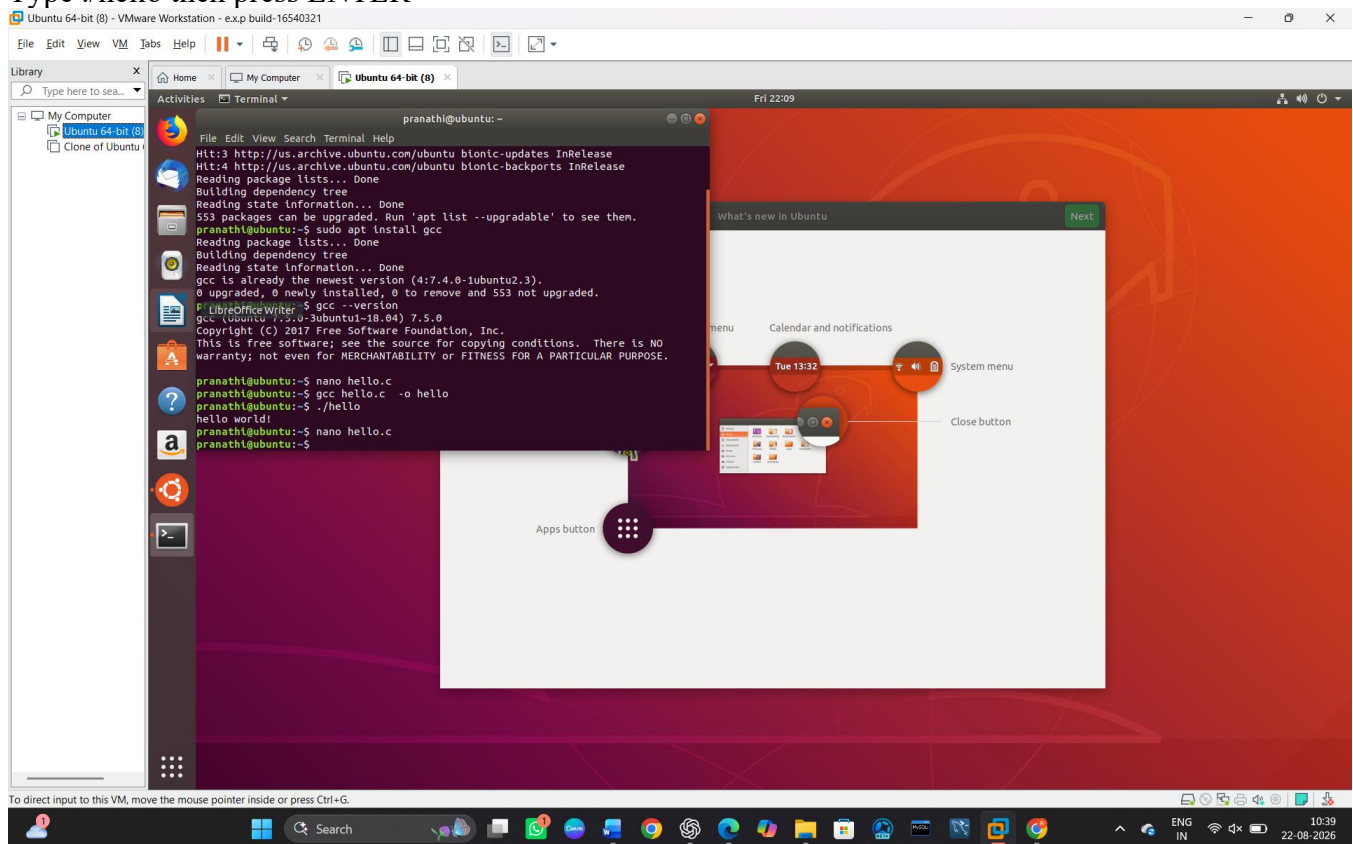
Step 5: Save the program

By ctrl O then press Enter Then Ctrl X..will return to terminal



Step 6: Run the program

Type ./hello then press ENTER



Result:

The C compiler was successfully installed in the Ubuntu virtual machine running on VMware Workstation. A simple C program was successfully compiled and executed, and the expected output **“Hello World!”** was obtained.